

SUMMARY OF CHAPTER 1

INTRODUCTION TO LANGUAGES

Language – way of communication between user and system

Types of Languages:

1. High Level Language: Refers to the programming languages such as java, python, c, c++. It is basically of English language to make it understandable for users to learn easily.
2. Middle Level Language: Includes the commands which is used in hardware such as Arduino, microprocessors, microcontrollers.
3. Low Level Language: refers to the byte code which is understandable by machines includes binary values 0,1. understand the input with the help of ASCII values. Finding ASCII values includes the process of taking LCM of a particular character value and finding its binary value such as 10010(for eg).

OVERVIEW OF JAVA

- Java introduced by James Gosling by sun microsystems.
- Later it has been bought by Oracle. Latest Java version is Java 26.
- Open- source platform.
- Basically, it is a high level programming language and fully object oriented programming language.
- Platform independent (Write Once Run Anywhere). Uses OOPS concepts. Contains Garbage Collection to reduce the memory for another program to run in JVM.
- Helps in developing applications supports as a memory management.

FEATURES OF JAVA

Scalability, Secure, Multithread, High Performance, Robust.

COMPILER IN JAVA

- It is used to convert the java code to byte code with the help of Java Virtual Machine (JVM).
- Checks the syntax. During compilation .class file is created which is located in bin folder.

INTERPRETER IN JAVA

- It is used to run the compiled java code which is bytecode. Reads line by line.
- Converts byte code to machine code. Use Just In Time Compiler to improve the performance.
- Find errors line by line.

WORKFLOW

Java file(.java file) -> compiler -> Byte code(.class file created) -> interpreter with JIT -> Machine code(understandable by machines)

JVM (JAVA VIRTUAL MACHINE)

- It loads the byte code verify them checking of syntax executes the code with the help of JIT. Both compilation and interpretation will be done.
- It runs garbage collector to free up the unused memory while execution.

JRE (JAVA RUNTIME ENVIRONMENT)

- It comprises JVM and primarily checks for packages, libraries and their supporting files.
- It provides environment where java application can run. Not for compilation
- JDK is used for modify or compile the code
- JRE= JVM+ required libraries

JDK (JAVA DEVELOPMENT KIT)

- It allows to write, compile, run the code
- Without JDK we can run the java program using JRE.
- For compilation and modification of code JDK is required.

IDE (INTEGRATED DEVELOPMENT ENVIRONMENT)

- It creates a platform to write the code for users.
- IDE's includes:
 - Eclipse
 - IntelliJ IDEA
 - NetBeans
 - VS Code
- It provides environment for write, compile, debug and run the code.

FEATURES

- Editing the code
- Compilation of java code to byte code
- Interpretation of byte code to machine code
- Debugging of code and fix errors
- Organize files, libraries

JAVA MAGIC BYTECODE

- It serves like an ID for all java programs and checks if the particular program is java class file.
- Every .class file starts with CAFEBABE with the help of this it checks all the files for execution.
- JVM opens .class file, checks if starts with CAFEBABE
- Rejects it if it doesn't contain magic code.
- Also checks if it is a secure program for execution.

JIT (JUST IN TIME COMPILER)

- It runs with the interpreter to make the execution faster.

- Helps when the given data is huge and during multiple loops or method calls.
- It compiles the part of byte code to native code having loops.
- Then it runs the native code rather than interpreting again and again.

NATIVE MACHINE CODE

- It consists of binary (0,1).
- Understandable by machine
- Source code(.java) -> compiles(.javac) -> JVM (interpreter+JIT) -> Native machine code